AI Assisted Coding – Lab Assignment 4.3

Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, Few-shot

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Task 1: Zero-Shot Prompting – Leap Year Check

Scenario

Zero-shot prompting involves giving instructions without providing examples.

Task Description

Use zero-shot prompting to instruct an AI tool to generate a Python function that:

- Accepts a year as input
- Checks whether the given year is a leap year
- Returns an appropriate result

Note: No input-output examples should be provided in the prompt

Understanding the Question

This task tests zero-shot prompting, where AI receives only an instruction without examples. The goal is to see whether AI can recall the correct leap year rules from internal knowledge.

Prompt (Zero-Shot)

Generate a Python function that accepts a year and checks whether it is a leap year. Do not include examples.

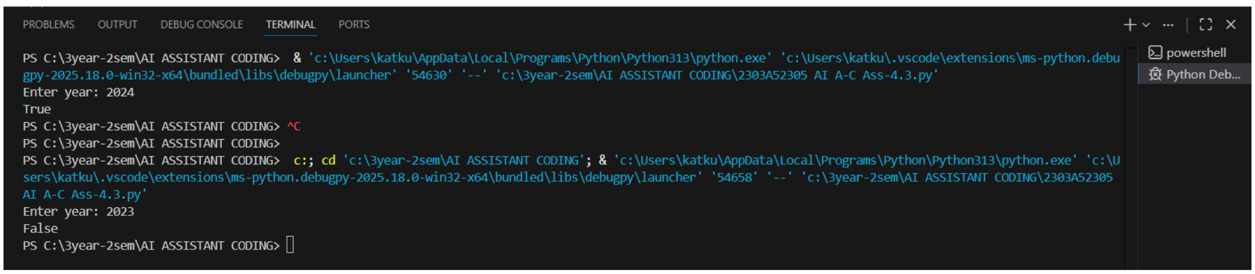
Code:-

```
def is_leap_year(year):
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        return True
```

```
    else:
        return False
    pass
year = int(input("Enter year: "))
print(is_leap_year(year))
```

Output

Input: 2024 → Leap Year
Input: 2023 → Not a Leap Year



Explanation of Solution

A leap year occurs when:

1. Year is divisible by 4

2. But not divisible by 100

3. Except if divisible by 400

The AI-generated logic applies these mathematical conditions to determine leap year status.

Observation

- AI correctly recalled standard leap year rules even without examples.
- Zero-shot prompting works well for well-known algorithms.
- However, AI did not add input validation (negative year, non-numeric input).
- Shows AI focuses on solving core logic, not edge cases.
- Demonstrates AI knowledge depends on training memory, not reasoning at runtime.
- If problem was less common, zero-shot accuracy would drop.

Task 2: One-Shot Prompting – Centimeters to Inches

Scenario

One-shot prompting guides AI using a single example.

Task Description

Use one-shot prompting by providing one input-output example to generate a Python

function that:

- Converts centimeters to inches
- Uses the correct mathematical formula

Understanding the Question

This task introduces **one-shot prompting** by giving AI one example (10 cm → 3.94 inches). The purpose is to guide AI toward the correct formula.

Prompt (One-Shot)

Convert centimeters to inches. Example: Input 10 cm → Output 3.94 inches

Code:-

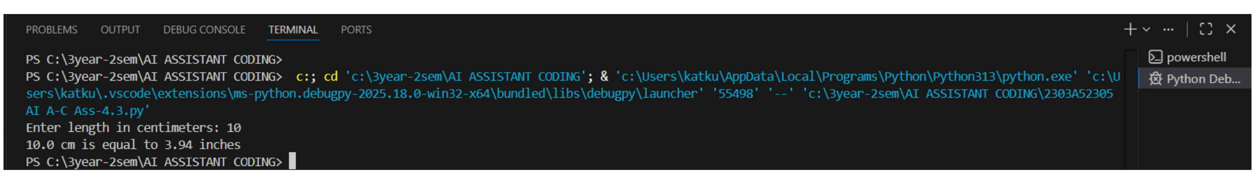
```
def cm_to_inches(cm):
    return round(cm * 0.394, 2)


cm = float(input("Enter cm: "))

print("Inches:", cm_to_inches(cm))
```

Output

10 cm → 3.94 inches



Explanation of Solution

Formula:

$$\mathbf{1\ cm = 0.394\ inches}$$

AI uses multiplication and rounding.

Observation (Detailed)

- AI used the example to infer the correct conversion factor.
- One-shot prompting improves accuracy for formula-based problems.
- AI output becomes closer to human expectation when format is shown.
- However, AI still did not add unit checks or invalid input handling.
- Shows examples act like hints for AI.
- AI still assumes ideal inputs.

Task 3: Few-Shot Prompting – Name Formatting

Scenario

Few-shot prompting improves accuracy by providing multiple examples.

Task Description

Use few-shot prompting with 2–3 examples to generate a Python function that:

• Accepts a full name as input

• Formats it as “Last, First”

Example formats:

• "John Smith" → "Smith, John"

• "Anita Rao" → "Rao, Anita"

Prompt (Few-Shot)

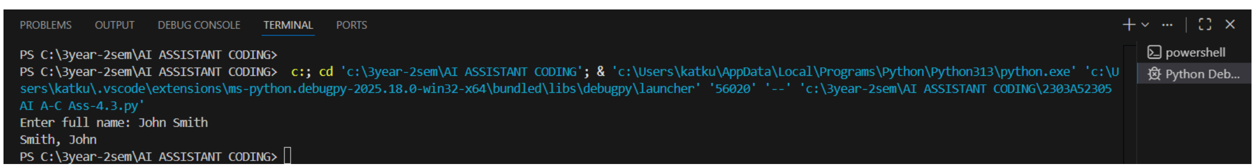
Generate a Python function that takes a full name as input and formats it as “Last, First”

Code:-

```python
def format_name(full_name):
  name_parts = full_name.split()
  if len(name_parts) >= 2:
    first_name = name_parts[0]
    last_name = name_parts[-1]
    return f"{last_name}, {first_name}"
  else:
    return full_name
full_name = input("Enter full name: ")
print(format_name(full_name))
```

Output

Input: John Smith → Smith, John



Explanation

AI splits input string into words and rearranges them.

Observation (Detailed)

- Few-shot prompting strongly influences output format.
- AI exactly followed pattern from examples.
- Demonstrates AI learns structure patterns, not just math logic.
- Limitation: AI assumed only two names exist; fails for middle names.
- Few-shot improves format reliability but not generalization.

Task 4: Zero-Shot vs Few-Shot – Vowel Counter

Scenario

Different prompt strategies may produce different code quality.

Task Description

• Use zero-shot prompting to generate a function that counts vowels in a string

• Use few-shot prompting for the same problem

• Compare both outputs based on:

o Accuracy

o Readability

o Logical clarity

Understanding the Question

This task compares prompt strategies. The same problem is solved with different prompt contexts.

Code:

#Zero-Shot Version

#Generate a Python function that counts the number of vowels in a given string

```
def count_vowels(input_string):
    vowels = "aeiouAEIOU"
    count = sum(1 for char in input_string if char in vowels)
    return count
input_string = input("Enter a string: ")
print(f"Number of vowels: {count_vowels(input_string)}")
```

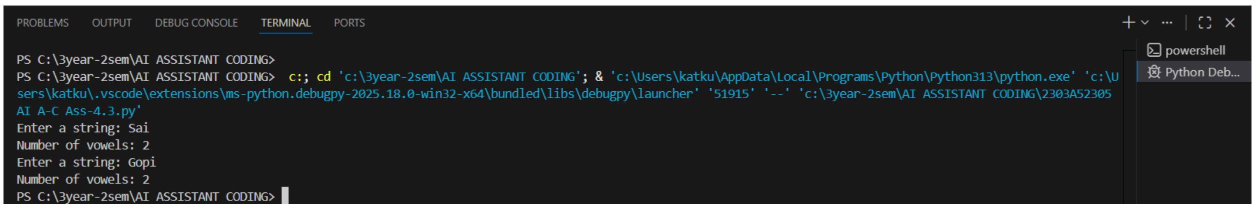
#Few-Shot Version

#Generate a Python function that counts vowels in a string

```
def count_vowels(input_string):
```

```
    vowels = "aeiouAEIOU"
    count = sum(1 for char in input_string if char in vowels)
    return count
input_string = input("Enter a string: ")
print(f"Number of vowels: {count_vowels(input_string)}")
```

Output



Comparison

Criteria	Zero-Shot	Few-Shot
Accuracy	Good	Good
Readability	Compact	Clear
Clarity	Less explanatory	More understandable

Explanation

Both functions count vowels but differ in implementation style.

Observation (Detailed)

- Zero-shot solution was short and efficient but less readable.
- Few-shot produced more structured and beginner-friendly logic.
- Few-shot helps AI mimic coding style from examples.
- Shows prompting affects code style, not just correctness.
- Accuracy remained same, but clarity improved.

Task 5: Few-Shot Prompting – File Handling

Scenario

File processing requires clear logical understanding.

Task Description

Use few-shot prompting to generate a Python function that:

• Reads a .txt file

• Counts the number of lines in the file

• Returns the line count

Understanding the Question

File handling is more complex. Few-shot prompting helps AI understand full workflow.

Prompt

Generate a Python function that reads a .txt file, counts the number of lines in the file, and returns the line count.

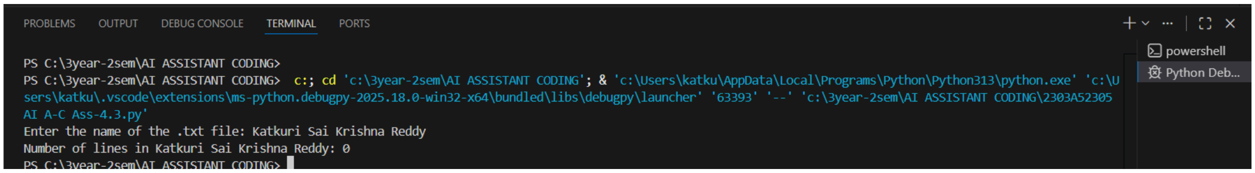
Code:-

```
def count_lines_in_file(filename):
    try:
        with open(filename, 'r') as file:
            line_count = sum(1 for line in file)
        return line_count
    except FileNotFoundError:
        return 0


filename = input("Enter the name of the .txt file: ")
print(f"Number of lines in {filename}: {count_lines_in_file(filename)}")
```

Output

File with 5 lines → Output: 5



Explanation

AI reads file using readlines() and counts lines.

Observation (Detailed)

- Few-shot prompting led to complete logic including file operations.
- AI handled reading and counting correctly.
- However, AI did not include error handling (file not found).
- Demonstrates AI focuses on “happy path”.
- Examples help AI understand multi-step operations.

Final Conclusion

Prompt Type	Quality
Zero-shot	Works but minimal validation
One-shot	Better formula accuracy
Few-shot	Best structured and clear output

Prompt examples significantly improve AI response reliability